TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Detection of Fabric Specification from Given Samples: Structure, Density, Stitch Length, Yarn Count, Weight and Fibre

Given an unknown knitted sample, produce its full working specification: identify the structure, count wales and courses, measure the stitch length and the yarn count from unravelled yarn, weigh the fabric by GSM cutter, and outline the fibre — every result in this sheet worked through on one real sample so the arithmetic can be checked line by line.

TE-3102 · KNITTING TECHNOLOGY SESSIONAL · STUDY 5 · LEVEL 3 TERM 1

1. Object and order of work

Object: to determine, from an unlabelled sample, every item of the specification needed to reproduce or judge the fabric, presented as one specification sheet (Section 9). Non-destructive tests come first, and each result feeds the next calculation:

1. **Identify the structure** by eye and lens (Section 2) — weft or warp, and which structure.
2. **Count wales and courses per cm** (Section 3), relaxed and flat.
3. **Measure the stitch length** from an unravelled course (Section 4).
4. **Find the yarn count** from unravelled yarn (Section 5) — its measured length and weight.
5. **Measure the weight (GSM)** by cutter and balance (Section 6), cross-checked against a GSM calculated from Sections 3–5 (Section 7).
6. **Identify the fibre** in outline (Section 8) — burning behaviour, microscopy, solubility.

Condition of the sample. All measurements are made on fabric **relaxed and flat**: laid on the bench without tension (dry-relaxed at minimum), away from the selvedge, the tube fold and any cut edge, where loops distort. Fabric measured straight off the roll, still stretched from batching, gives course and wale counts that are simply wrong — relaxation is the first test condition, and the viva will ask for it.

2. Identifying the structure

- **Find the directions first.** The **wales** run along the columns of loops, the **courses** across them. On the technical face of plain weft knitting the wales show as smooth vertical ribs of loop legs; the reverse shows the arcs (loop heads and sinker loops) as horizontal rows.
- **Weft or warp?** Tease out one edge yarn. A **weft** fabric unravels course by course — one yarn runs across the whole width and pulls out a complete row. A **warp** fabric does not: its yarns run lengthwise, each in its own wale with underlaps behind (Study 2), and an edge yarn removes only its own zig-zag path.
- **Which weft structure? Single jersey:** all loops to one side — face and reverse clearly different, edges curl. **Rib:** loops to both sides in alternating wales — faces alike, vertical cords, no curl, high crosswise elasticity. **Interlock:** smooth and alike both sides like rib but firmer, less extensible, no visible cord — two rib courses intermeshed. **Purl (links-links):** loops to opposite sides in successive *courses*. Confirm under the lens by the direction loop heads point — never by handle alone.
- **Tuck, miss and pattern.** Look for loops held over two or more courses (tuck), yarns floating behind wales (miss), transferred or crossed loops. Sketch a few wales and courses (Figure 1) as your record: the sketch *is* the identification the report is marked on.

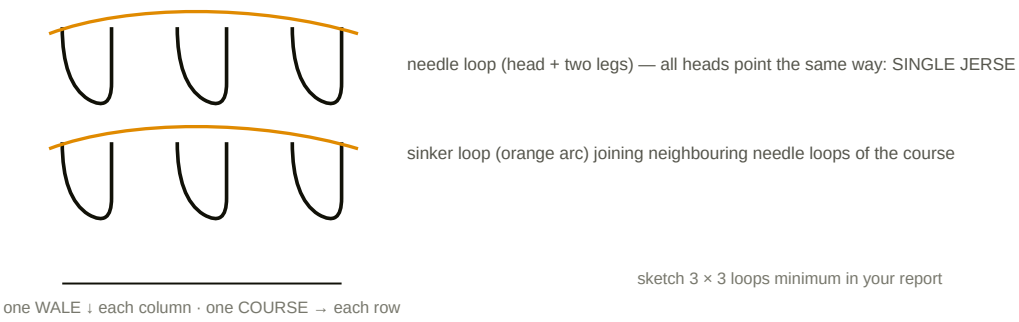


Figure 1 — The loop structure of single jersey on the technical face: needle loops in vertical wales, joined across each course by sinker loops. In rib, alternate wales would be drawn to the reverse; in warp fabric, each wale would carry its own yarn with underlaps behind.

3. Wales and courses per centimetre

Apparatus: counting glass (pick glass) or a lens with a ruled centimetre scale, and a needle pointer. Lay the relaxed sample flat, place the glass with its scale exactly across the wales, and count wales in 1 cm (count 2–3 cm and divide if loops are coarse); turn the sample and count courses per cm likewise. Count at **three separate places** and average — a single count is an anecdote, not a measurement. Avoid faults, creases and the tube fold.

> **WORKED RESULT — THE SAMPLE OF THIS SHEET**
>
> Counts at three places gave wales 13, 13, 13 per cm and courses 21, 20, 22 per cm. Averages: **WPC = 13 per cm; CPC = 21 per cm**. Stitch density = WPC × CPC = 13 × 21 = **273 loops per**

cm². (For the report in inches: 13 × 2.54 ≈ 33 wales per inch; 21 × 2.54 ≈ 53 courses per inch.)

What the counts mean. Wales per cm reflects machine gauge and widthways relaxation; courses per cm reflects course spacing, set jointly by stitch length and take-down (Study 3). With stitch length they fix tightness: the same counts at a longer stitch length give a looser, lighter fabric — a specification is never complete with counts alone.

4. Stitch length by unravelled yarn

Method. Mark two points exactly **100 wales** apart along one course (lens count; contrasting thread or pen dots). Unravel that course between the marks, straighten the yarn **without stretching it** — along the rule, or under the light standard tension your laboratory specifies, just enough to remove the loop crimp — and measure its length. Then:

$$\text{stitch length (mm)} = \text{length of yarn from 100 loops (mm)} \div 100$$

WORKED RESULT — STITCH LENGTH

The unravelled course from 100 marked wales measured **340 mm** straightened. Stitch length = 340 ÷ 100 = **3.4 mm**. Repeat on two further courses and average; the three values should agree within a few hundredths of a millimetre on a well-set machine — a course that differs markedly is the signature of the feeder fault behind barré (Studies 3 and 4).

Errors to avoid: stretching the yarn while measuring (commonest error — it inflates stitch length and every calculation built on it); miscounting the wales between marks; yarn taken too near a selvedge.

5. Yarn count from the crimped (unravelled) yarn

Method. Unravel yarn from the sample (Section 4's courses plus more as needed), measure its total straightened length as in Section 4, and weigh the whole of it. Count follows from length and weight — always the **straightened unravelled length**, never the fabric length it came from.

WORKED RESULT — YARN COUNT (VERIFY EACH LINE)

Yarn unravelled from the sample: total straightened length **30.0 m**, weight **0.591 g**.

In tex (grams per 1,000 m): tex = 0.591 × (1,000 ÷ 30.0) = 0.591 × 33.33 = **19.7 tex**.

In cotton count Ne: Ne = 590.5 ÷ tex = 590.5 ÷ 19.7 = 29.97 ≈ **30 Ne**.

So the sample is knitted from (nominally) 30 Ne yarn — the same count as the laboratory machine of Study 1, as it should be: the sample is that machine's fabric, and this test is how you would

prove it from the fabric alone. Report both systems, and state the result as “19.7 tex (≈ 30 Ne), from unravelled yarn” so the method travels with the number.

Accuracy note: unravelled yarn has lost a little fibre in knitting and handling; agreement within about half a count of the nominal value is a good result by this standard method.

6. Fabric weight — GSM by GSM cutter

Apparatus: the GSM cutter (circular cutter, disc of exactly **100 cm²**) and a balance reading to 0.01 g. Lay the relaxed fabric flat on the mat, away from edges and faults, cut with a firm rotary action, and weigh the disc. The disc is one-hundredth of a square metre, so:

$$\text{GSM (g/m}^2\text{)} = \text{weight of the 100 cm}^2\text{ disc (g)} \times 100$$

WORKED RESULT — GSM

The 100 cm² disc from the sample weighed **1.85 g**. GSM = 1.85 × 100 = **185 g/m²**. Cut and weigh **three discs** from different parts of the sample and average them; report the average and the spread — a wide spread across one sample is itself a finding (uneven feeding or take-down, Studies 3–4).

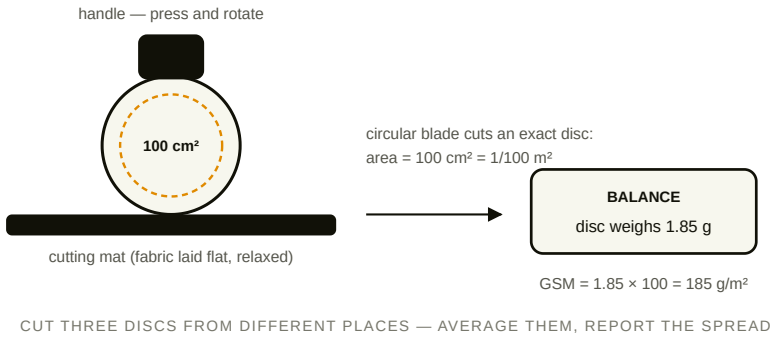


Figure 2 — GSM determination: the cutter takes a disc of exactly 100 cm², so the disc’s weight in grams, multiplied by 100, is the fabric weight in g/m².

7. Cross-check: GSM calculated from the other results

The measurements of Sections 3–5 must predict Section 6: one square metre contains (WPC × 100) × (CPC × 100) loops, each holding a stitch length of yarn of the measured count. In tex:

$$\text{GSM (calculated)} = \text{WPC} \times \text{CPC} \times \text{stitch length (mm)} \times \text{tex} \div 100$$

WORKED CROSS-CHECK — THE SAMPLE OF THIS SHEET

GSM = 13 × 21 × 3.4 × 19.7 ÷ 100 = 273 × 3.4 × 19.7 ÷ 100 = 18,285 ÷ 100 ≈ **183 g/m²**.

Measured by cutter: **185 g/m²**. Agreement is within about 1% — an excellent cross-check. The

small excess of the measured value is expected: the cutter method weighs the fabric as it is, including its moisture regain and any finish, while the calculation counts only yarn. **If your two values ever differ by more than a few per cent, a measurement is wrong** — re-check the stitch length for stretching first, then the counts, then the weighing; the cross-check exists precisely to catch this before the report is written.

8. Fibre identification — outline

Only an outline is required at this level; full fibre analysis belongs to the testing course. Work in this order and record what you actually observe:

- **Handle and appearance:** lustre, warmth, crease recovery — suggestive only, never conclusive.
- **Burning behaviour:** yarns held in tweezers to a flame edge. Cotton approaches without melting, burns quickly, smells of burning paper, leaves soft grey ash. Synthetics shrink and melt from the flame to a hard bead (polyester: hard dark bead, sweetish smell; nylon: hard pale bead). Wool/silk burn slowly, self-extinguish, smell of burning hair, leave a crushable dark bead. Record approach, flame behaviour, odour and residue as four separate observations.
- **Microscopy:** yarn teased out on a slide — cotton a twisted ribbon with central lumen, wool scaled, synthetics smooth uniform rods. One clear longitudinal feature outweighs a paragraph of burning notes.
- **Solubility (confirmation; instructor present):** the testing course's standard tests — e.g. cotton dissolving in concentrated sulphuric acid, wool in hot sodium hydroxide, polyester resisting both. State test and result; never improvise reagents.
- **Blends:** if burning and microscopy disagree, suspect a blend and say so — "cotton-rich blend, synthetic component present" is a correct answer where a confident wrong fibre is not.

9. The specification sheet — the finished result

ITEM	RESULT FOR THE SAMPLE OF THIS SHEET
Structure	Weft-knitted single jersey (plain); face/reverse identified, sketch Figure 1
Wales per cm / Courses per cm	13 per cm / 21 per cm (averages of three counts, relaxed sample)
Stitch density	273 loops per cm²
Stitch length	3.4 mm (unravelled course, 100 wales, 340 mm)
Yarn count	19.7 tex ≈ 30 Ne (30.0 m unravelled yarn, 0.591 g)

Fabric weight	185 g/m² by cutter (1.85 g disc); 183 g/m² calculated — agreement within about 1%
Fibre	By burning test and microscopy: cotton (state your own observations for your sample)
Reproduction note	Knittable on the Study 1 machine (24 gauge) at stitch length 3.4 mm in 30 Ne yarn; its sheet set 3.0 mm — the longer loop here gives the heavier, denser fabric measured

10. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. A 100 cm² disc weighs 1.85 g. State the GSM, and the two reasons for cutting three discs, not one.
2. An unravelled course from 100 wales measures 340 mm, and 30.0 m of unravelled yarn weighs 0.591 g. Give the stitch length and the yarn count in tex and Ne, showing the working.
3. From WPC 13, CPC 21, stitch length 3.4 mm and 19.7 tex, calculate the expected GSM; comment on cutter results of 185 and of 150 g/m².
4. Your burning test suggests cotton, but a melting bead is also present. Conclusion, and next test?

Answer key

1. GSM = 1.85 × 100 = **185 g/m²**. Three discs because (i) one may land on a fault or thin place, and (ii) the *spread* between discs is itself information — the evenness of feeding and take-down.
2. Stitch length = 340 ÷ 100 = **3.4 mm**. Tex = 0.591 × (1,000 ÷ 30.0) = **19.7 tex**; Ne = 590.5 ÷ 19.7 ≈ **30 Ne**.
3. Expected GSM = 13 × 21 × 3.4 × 19.7 ÷ 100 ≈ **183 g/m²**. A cutter result of 185 agrees within about 1% — accept both (the excess is moisture and finish). A result of 150 disagrees by some 18% — a measurement is wrong: re-check stretched yarn first, then counts, then weighing.
4. A **blend**: cellulosic (cotton-like burning) plus a thermoplastic synthetic (the melting bead). Next: **microscopy** — two fibre populations (ribbons and smooth rods) — then solubility separation if a ratio is required.